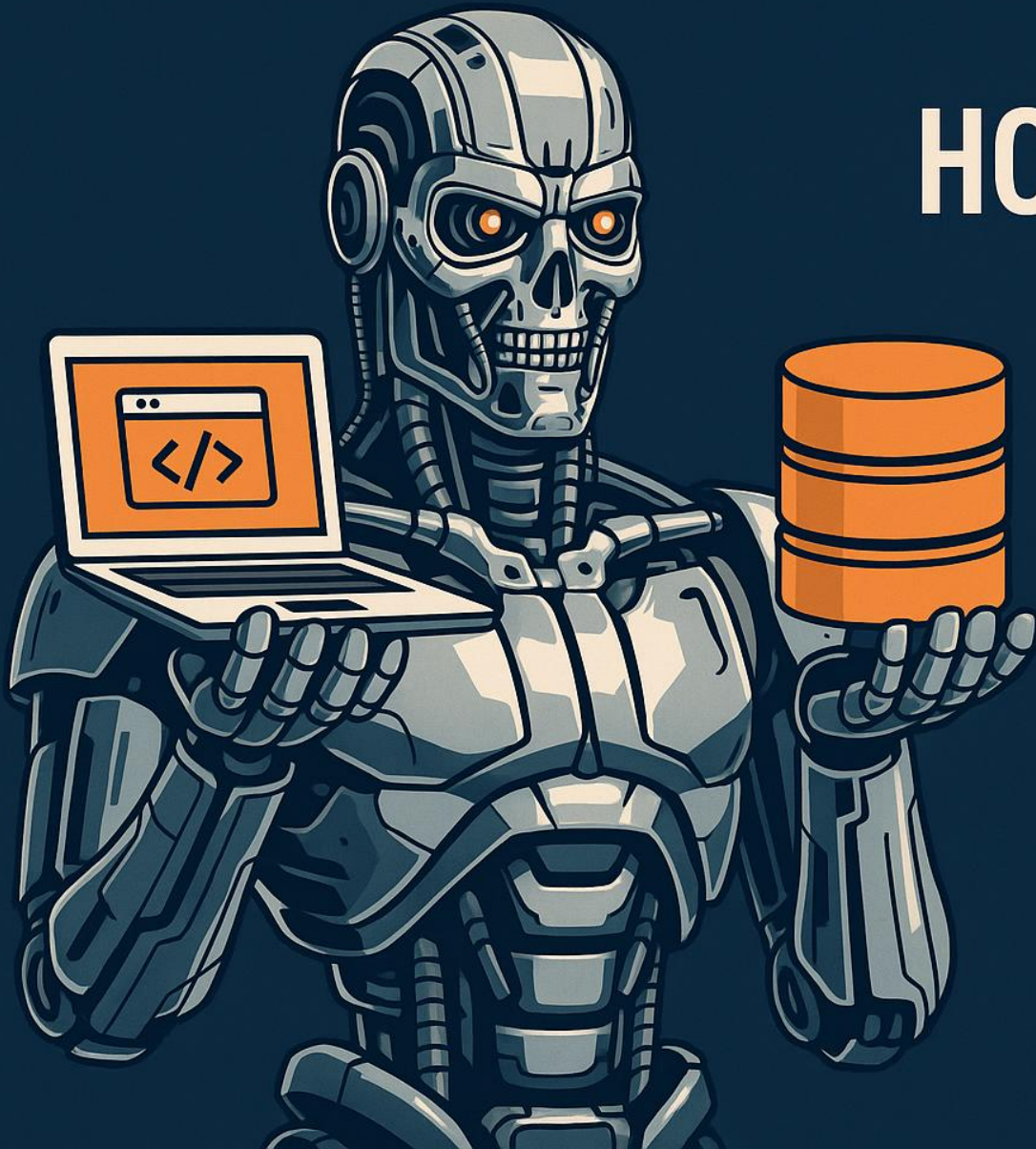




HOSTING APPS AND USING DATABASES

Overview

- **Hosting apps online**
- **Add a database to an app**
- **MCP servers**

HOSTING APPS ONLINE

Hosting App Online

- **First, push your app code to GitHub**
- **Once the app is on GitHub you can host it online**
- **Hosting means pay someone to run your app on their server \$\$\$**

Hosting Services

- **There are many hosting providers**
- **App hosts – simple to use, good for demos**
 - Render
 - Vercel
- **Hyperscalers – large scale apps**
 - Amazon Web Service (AWS)
 - Microsoft Azure
 - Google Cloud Platform (GCP)

App Hosts

- **Vercel**
 - Hosts frontends, like React
 - Does not host backends, like long running agents
 - Generous for frontends: web service does not sleep quickly
- **Render**
 - Hosts front ends, like React
 - Hosts backends
 - Not so generous: web services sleep after idle

Hosting a React App on Render

- We will use Render to host React Apps
 - Go to <https://render.com>
 - Click "Sign Up" and choose GitHub as login method
 - Authorize Render to access your GitHub repos
- Render will update the app automatically each time we push to the repo

Create New Web Service

- Our app will be a new Static Site
- This information will be auto-filled by Render
 - The branch might be master instead of main

1. Click "New" → "Static Site"
2. Select your repo: `flag-time`
3. Fill out deployment info:
 - Name: `flag-time`
 - Branch: `main`
 - Build Command: `npm run build`
 - Publish Directory: `build`
 - Click "Create Static Site"

Let Render Cook

- Render will:
 - Install dependencies (`npm install`)
 - Run `npm run build` to generate production files
 - Host from the `build/` directory
- It takes ~1 minute to deploy
- ✓ Once done, you'll get a live URL (e.g., `https://flag-time.onrender.com`)

Deployed App

 STATIC SITE

flag-time

Manual Deploy ▾

 zlisto / flag-time  master

<https://flag-time.onrender.com> 

July 17, 2025 at 9:25 PM ✓ Live

[120610c](#) Add flags for Bangladesh, Korea, and France; update country list in App.js

All logs ▾

 Search

 Live tail ▾

GMT+7

Jul 17 09:26:58 PM 

Jul 17 09:26:58 PM  The project was built assuming it is hosted at /.

Jul 17 09:26:58 PM  You can control this with the homepage field in your package.json.

Jul 17 09:26:58 PM 

Jul 17 09:26:58 PM  The build folder is ready to be deployed.

Jul 17 09:26:58 PM  You may serve it with a static server:

Jul 17 09:26:58 PM 

DATABASES

Databases

- **A database is a structured place to store, access, and manage data**
- **Apps need to remember things**
 - Users & passwords
 - AI chat history
 - Product information
- **Without a database**
 - You lose everything when the app restarts
 - No sharing of data across users or sessions

Database Architecture

- In a database hosting site, you will create a **project** which comes with a **database**
- Inside the database you create **tables**
 - Your app will read and write data from the tables
- Table examples
 - Users: has username and encrypted passwords
 - Chats: has chat history
 - Products: has item names, prices, and other metadata

Supabase

- **Supabase is an easy to use cloud database**
 - Uses Postgres databases
 - <https://supabase.com/>
- **Easy to connect to your React App**
- **Supabase also has file storage**
 - You can store images, videos, etc
 - Databases store text (and sometimes images if you are clever)

MCP SERVERS

MCP Servers

- **Many tasks we do to host an app can be done by an AI agent**
- **This is done through the platforms' MCP servers**
 - **Vercel, Render, Supabase**
- **MCP – model context protocol**
 - **A server with instructions and API connections for any AI agent**

MCP Servers Background

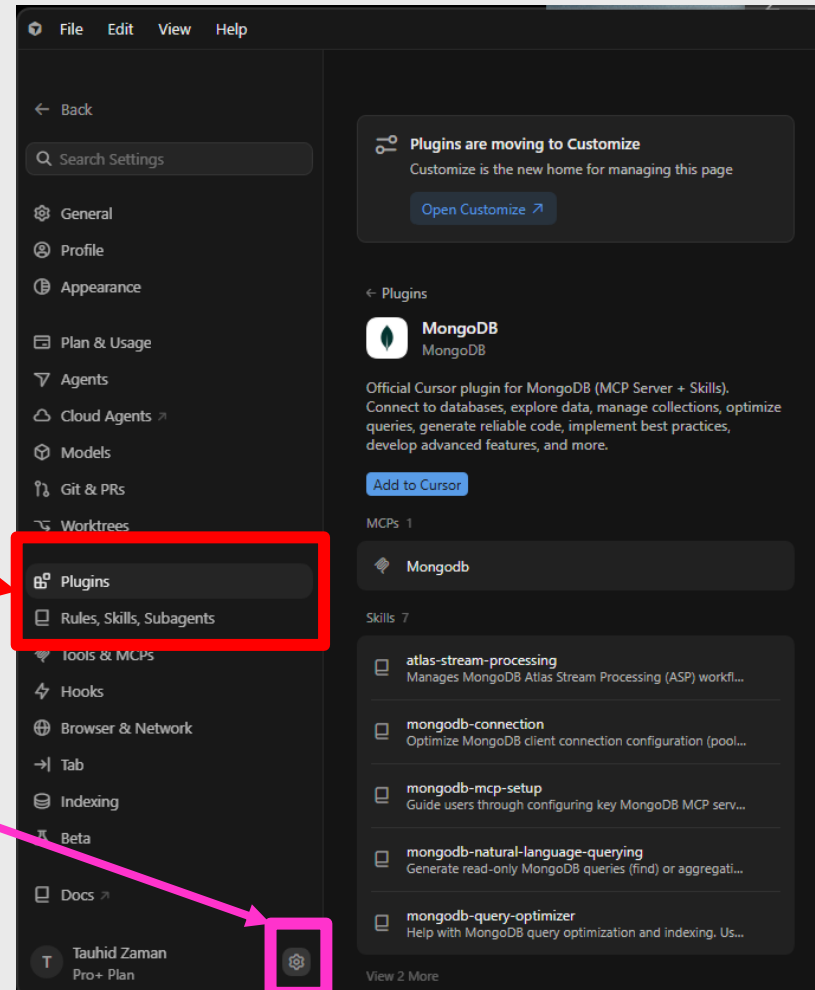
- **Anthropic created MCP in late 2024**
- **AI was getting good at writing code**
- **AI was not good at doing things in the real world**
 - **Provisioning a database, deploy an app, read live database logs**
 - **Required custom code to perform actions**
- **MCP servers provides a set of instructions and tools for AI vibe coders**
- **Major platforms adopted MCP within months**

Connecting to an MCP Server

- You can ask your AI vibe coder to connect you to the MCP server of a platform
 - Ex) connect me to the Vercel MCP server
- The agent will do what it must do to connect to the server
 - You will have to authenticate the connection
- In Cursor, you can control the MCP server **scope**
 - Global – all your agents can use them
 - Workspace only – only agents in a specific workspace can use them

MCP Server Plugins

- Some MCP servers have plugins
 - Vercel, Render, Supabase
- You can install them via **Plugins**
- Plugins is in the **Settings**



Using an MCP Server

- Once you are connected and authenticated on an MCP server, your agent can do everything for you
- Ex: “yo check my supabase mcp server and tell me my table names”

Connected to  <https://qscblodjgsfckmcscivx.supabase.co>

Tables:

Table	Rows
users	0
courses	1
roster_entries	774
attendance_events	4
attendance_records	155